Logic or Propositional Calculus
and
First-order Predicate Calculus

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Logic or Propositional Calculus (1/3)

Logic or propositional calculus is based on statements, which have **truth values** (true or false).

A proposition, or statement, is any declarative sentence, which is either true (T) or false (F).

We refer to **T** or **F** as the truth value of the statement.

The calculus provides a means of determining the truth values associated with statements formed from "atomic" statements.

Logic or Propositional Calculus (2/3)

If p stands for "pressure is high in pipe P_1" and q for "pipe P_1 is leaking" then we may form statements such as shown in table below

Table 1-1. Symbolic Statements

Symbolic Statement	Translation
$p \vee q$	p or q
$p \wedge q$	p and q
$p \Rightarrow q$	p logically implies q
$p \Leftrightarrow q$	p is logically equivalent to q
$\neg p$ (also $\sim p$)	Not p

Logic or Propositional Calculus (3/3)

Note that $\vee$, $\wedge$, $\Rightarrow$, $\Leftrightarrow$, are all binary connectives. They are sometimes referred to, respectively, as the symbols for ***disjunction***, ***conjunction***, ***implication*** and ***equivalence***.

Also $\neg$ is unary and is the symbol for negation.

First-order Predicate Calculus (1/17)

- ***Propositional logic*** provide us with the means to assess the truth value of compound statements from the truth values of the "building blocks"
- ***Rules*** are required to do this.
- **For example**, the calculus states that "p ˅ q" is true if either p is true or q is true (or both are true).
- Similar rules apply for all the ways in which the building blocks can be combined. The language of predicate calculus requires:
  - Variables and Constants

First-order Predicate Calculus (2/17)

The language of predicate calculus requires:

Variables and Constants

Symbol	Meaning
$\vee$	or
$\wedge$	and
$\neg$	not
$\Rightarrow$	logically implies
$\Leftrightarrow$	logically equivalent
$\forall$	for all
$\exists$	there exists

First-order Predicate Calculus (3/17)

Quantification is non-logical constants that include names and entities.

For example:

$\forall$ x.man(x) $\Rightarrow$ mortal(x), means all men are mortal.

$\exists$ x.Tank(x), means there is at least one tank.

First-order Predicate Calculus (4/17)

- It is possible to form a new proposition from old one.
- For example, p: "There is Pump with 300 rpm in Plant Model Plant-1."
- The negation of p is ¬p, which is defined as: "There is no Pump with 300 rpm in Plant Model Plant-1."
- Another example: if p: "$1 + 4 < 5$", q: "$1 + 4 = 5$", then $\sim p \wedge \sim q$: "$1 + 4 > 5$".

First-order Predicate Calculus (5/17)

- ***Truth table*** provides basic definition of predicate calculus.

(a) Negation

p	$\neg p$
T	F
F	T

(b) Disjunction

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

(c) Conjunction

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

First-order Predicate Calculus (6/17)

There are set of more complex expressions, that can be further simplified to validate the underlying system. Conditional and bidirectional are other forms of predicate calculus. For example, if p, then q can also be represented as: p implies q, and we write $p \Rightarrow q$. This can be represented as $(\neg p) \vee q$. Table 1-4 shows the truth table of such logic.

Table . $(\neg p) \vee q$.

p	q	$p \Rightarrow q$	$\neg p$	$(\neg p) \vee q$
T	T	T	F	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

First-order Predicate Calculus (7/17)

Table . Examples of Conditional Statements

If p, then q. p implies q.
q follows from p. Not p unless q.
q if p. p only if q.
Whenever p, q. q whenever p.
p is sufficient for q. q is necessary for p.
p is a sufficient condition for q. q is a necessary condition for p.

First-order Predicate Calculus (8/17)

The biconditional $p \Leftrightarrow q$, which we read "p if and only if q" or "p is equivalent to q," is defined by the truth table, shown in table .

Table . Biconditional Statements.

p	q	$p \Leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

First-order Predicate Calculus (9/17)

Modus Ponens or Direct Reasoning can be defined as: $[(p \Rightarrow q) \wedge p] \Rightarrow q$, which can be represented as:

$p \Rightarrow q$ (if p then q, or p implies q)
p (p, which is true or false)

$\therefore q$

Similarly,
$(p \wedge q) \Rightarrow (r \wedge s)$
$(p \wedge q)$

$\therefore (r \wedge s)$

First-order Predicate Calculus (10/17)

Modus Tollens or Indirect Reasoning can be defined as:

$[(p \Rightarrow q) \wedge \neg q] \Rightarrow \neg p$

In words, if p implies q, and q is false, then so is p.

$p \Rightarrow q$

$\neg q$

$\therefore \neg p$

First-order Predicate Calculus (11/17)

Disjunctive Syllogism or One-or-the-Other:

$[(p \vee q) \wedge (\neg p)] \Rightarrow q$

$[(p \vee q) \wedge (\neg q)] \Rightarrow p$

Distributive Laws:

$p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r)$

$p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r)$

First-order Predicate Calculus (12/17)

Applying Modus Ponens

1. $(p \vee q) \Rightarrow (r \wedge \neg s)$
2. $\neg r \Rightarrow s$
3. $p \vee q$

Statement $(p \vee q)$ appears twice in lines (1) and (3). Looking at Modus Ponens, we see that we can deduce $(r \wedge \neg s)$ from these lines. Thus, we can enlarge our list as follows:

1. $(p \vee q) \Rightarrow (r \wedge \neg s)$	Premise
2. $\neg r \Rightarrow s$	Premise
3. $p \vee q$	Premise
4. $r \wedge (\neg s)$	1,3 Modus Ponens

First-order Predicate Calculus (13/17)

Summary of Rules of Inference

- T1 Any tautology that appears on the list at the end of the last section can be used as a rule of inference.
- T2 We can add any tautology that appears in the list of tautologies at the end of the last section as a new line in our list of true statements.
- S (Substitution): We can replace any part of a compound statement with a tautologically equivalent statement.
- C (Conjunction): If A and B are any two lines in a proof, then we can add the line AB to the proof.
- P (Premise): We can write down a premise as a line in a proof.

First-order Predicate Calculus (14/17)

Predicates

Upper case Roman letters, plus square brackets:
Examples:
H[a] : a is happy
R[a, b] : a respects b
S[a,b,g] : a sold b to g
H[a,b,g,d] : a is happy that b sold g to d

First-order Predicate Calculus (15/17)

Function Signs

Lower case Roman letters, plus parentheses:
Examples:
m(a) : the mother of a
s(a,b) : the sum of a and b
s(a,b,g) : the sum of a, b, g
Variables: lower case Roman letters: z, y, x, …

First-order Predicate Calculus (16/17)

- Logic is the study of good reasoning - and good reasoning is of considerable importance in many subjects.

- Elementary logic covers certain aspects of logic, which are regarded as fundamental.

In elementary logic, compound terms, such as x + y, are not used where the focus is on the part of logic concerned with purely logical rules rather than with rules deriving from mathematical practice, or from some other specific domain. This restriction enables the use of a fairly simple syntax, and to make certain parts of the underlying logic practice fairly efficient.

First-order Predicate Calculus (17/17)

In general, elementary logic can include the following:

- sentences S
- noun phrases N
- predicates $Nk \rightarrow S$
- function signs $Nk \rightarrow N$
- connectives $Sk \rightarrow S$
- quantifiers $V+S \rightarrow S$
- description operator $V+S \rightarrow N$

Specification techniques

- **Algebraic approach**
  - The system is specified in terms of its operations and their relationships
- **Model-based approach**
  - The system is specified in terms of a state model that is constructed using mathematical constructs such as sets and sequences. Operations are defined by modifications to the system's state.

    Also referred as Model checking. Model checking can be viewed as exhaustive simulation, i.e. exhaustively exploring the system state space to prove certain correctness properties

Formal specification Languages

	Sequential	Concurrent
Algebraic	Larch (Gutting et al., 1993) OBJ (Futatsugi et al., 1985)	Lotos (Bolognesi and Brinksma, 1987) FSP (Finite State Process)
Model-based	Z (Spivey, 1992) VDM (Jones, 1980) B (Wordsworth, 1996)	CSP (Hoare, 1985) CCS (Milner, 1980) Petri Nets (Peterson, 1981) LTS (Magee and Kramer, 2006) π-ADL (Flavio et al., 2004) UPPAAL

CSP (Communicating Sequential Processes)
CCS (Calculus of Communicating Systems)

Use of formal specification

- Formal specification involves investing more effort in the early phases of software development
- This reduces requirements errors as it forces a detailed analysis of the requirements
- Incompleteness and inconsistencies can be discovered and resolved
- Hence, savings as made as the amount of rework due to requirements problems is reduced